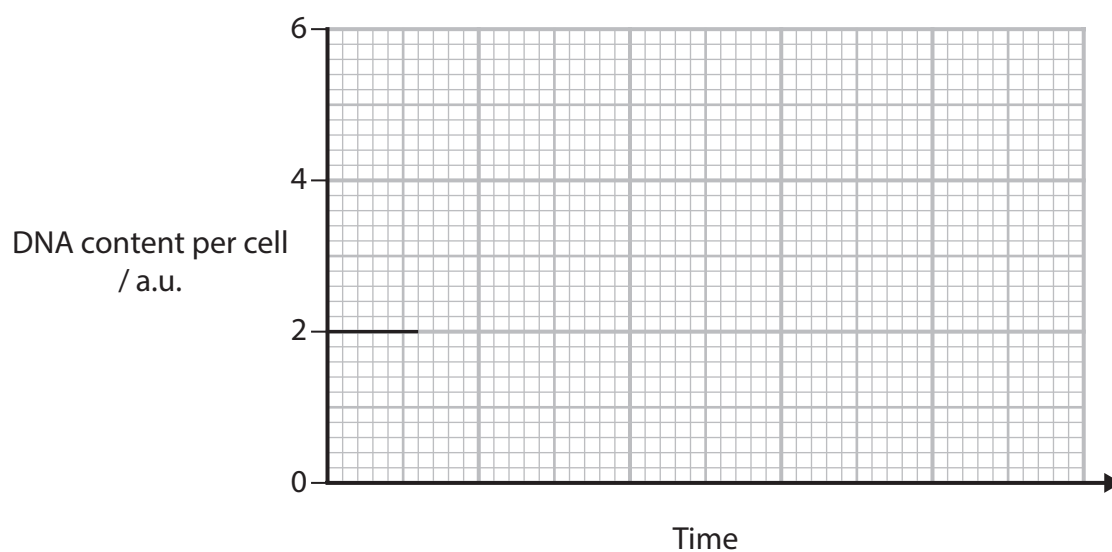


5 Some cells divide by meiosis to form gametes.

- (a) Complete this graph to show how the DNA content per cell would change when a cell, with 2 a.u. of DNA, undergoes meiosis to produce gametes.

(3)



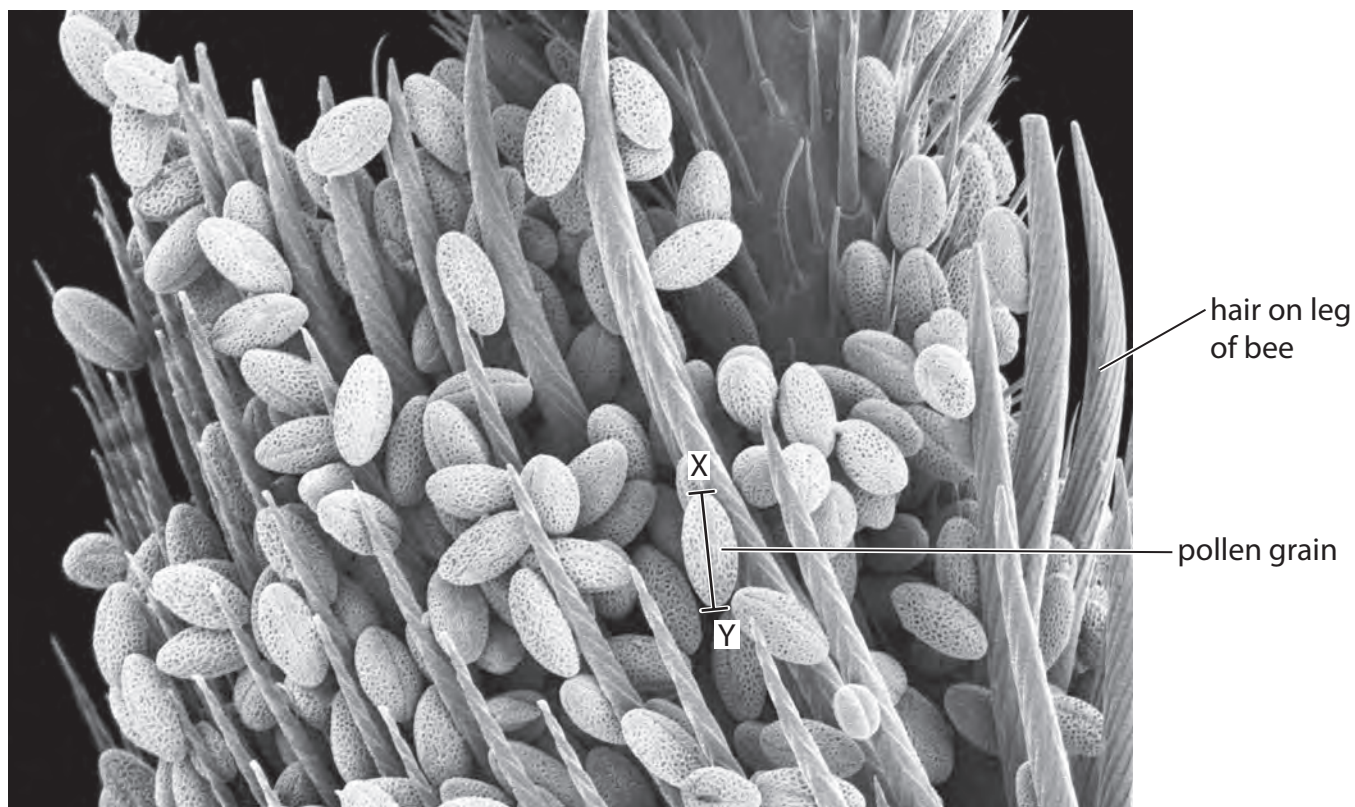
DO NOT WRITE IN THIS AREA

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(b) The photograph shows pollen grains on part of a leg of a bee.



(Source: © Susumu Nishinaga / Science Photo Library)

The actual length of the pollen grain between X and Y is $24\text{ }\mu\text{m}$.

Calculate the magnification of this photograph.

Give your answer to **two** significant figures.

(2)

Answer

- (d) The genetic diversity in a population of one species of peony was investigated.

There were 418 peony plants in this population.

Some of the genes of peony have several alleles.

The table shows the number of different alleles and the heterozygosity index for each of four genes in this population.

Gene	Number of different alleles of the gene	Heterozygosity index
PS43	10	0.422
PS103	10	0.810
PS105	4	0.036
PS147	16	0.834

- (i) How many of the peony plants in this population are heterozygous for gene PS105?

(1)

- ☐ A 15
- ☐ B 60
- ☐ C 340
- ☐ D 350

- (ii) The width of peony leaves is an example of polygenic inheritance.

Which row is correct for the width of peony leaves?

(1)

	One gene determines leaf width	Leaf width is determined by multiple alleles for several genes	Leaf width shows continuous variation
☐ A	✓	✓	✗
☐ B	✓	✗	✓
☐ C	✗	✓	✓
☐ D	✗	✗	✗

(Total for Question 5 = 12 marks)